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 Studierichting: Informatica
 Oefeningenreeks 1C nr. 48.1

Opgave: Bereken $(1+i)^n - (1-i)^n$, met $n = 2025$

$$(1+i)^{2025} - (1-i)^{2025}$$

$$z_1 = 1 + i$$

$$\Rightarrow r = \sqrt{a^2 + b^2} = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\Rightarrow \tan \theta = \frac{b}{a} = \frac{1}{1} = 1 \Rightarrow \theta = \text{Bgtan}(1) = \frac{\pi}{4}$$

$$\Rightarrow z_1 = r \text{ cis } \theta = \sqrt{2} \text{ cis } \left(\frac{\pi}{4} \right)$$

$$z_2 = 1 - i$$

$$\Rightarrow r = \sqrt{a^2 + b^2} = \sqrt{1^2 + (-1)^2} = \sqrt{2}$$

$$\Rightarrow \tan \theta = \frac{b}{a} = -\frac{1}{1} = -1 \Rightarrow \theta = \text{Bgtan}(-1) = -\frac{\pi}{4}$$

$$\Rightarrow z_2 = r \text{ cis } \theta = \sqrt{2} \text{ cis } \left(-\frac{\pi}{4} \right)$$

$$\stackrel{\text{De Moivre}}{\Rightarrow} (r \text{ cis } \theta)^n = r^n \text{ cis } (n\theta)$$

$$\left(\sqrt{2}^{2025} \text{ cis } \left(2025 \frac{\pi}{4} \right) \right) - \left(\sqrt{2}^{2025} \text{ cis } \left(-2025 \frac{\pi}{4} \right) \right)$$

$$= \sqrt{2}^{2025} \left(\text{cis} \left(2025 \frac{\pi}{4} \right) - \text{cis} \left(-2025 \frac{\pi}{4} \right) \right)$$

$$= \sqrt{2^{2024}} \cdot 2 \left(\text{cis} \left(2024 \frac{\pi}{4} + \frac{\pi}{4} \right) - \text{cis} \left(-2024 \frac{\pi}{4} - \frac{\pi}{4} \right) \right)$$

$$= 2^{1012} \cdot \sqrt{2} \left(\text{cis} \left(\frac{\pi}{4} \right) - \text{cis} \left(-\frac{\pi}{4} \right) \right)$$

$$= 2^{1012} \cdot \sqrt{2} \left(\cos \left(\frac{\pi}{4} \right) + i \cdot \sin \left(\frac{\pi}{4} \right) - \cos \left(-\frac{\pi}{4} \right) - i \cdot \sin \left(-\frac{\pi}{4} \right) \right)$$

$$= 2^{1012} \cdot \sqrt{2} \left(\frac{\sqrt{2}}{2} + i \cdot \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} + i \cdot \frac{\sqrt{2}}{2} \right)$$

$$= 2^{1012} \cdot \sqrt{2} \cdot (i\sqrt{2}) = i \cdot 2^{1012} \cdot \sqrt{2} \cdot 2 = i \cdot 2^{1012} \cdot 2$$

$$= 2^{1013} \cdot i$$

References